

Problem 1) system is isolated.

$$F = \frac{dP}{dt} \Rightarrow P(t_2) - P(t_1) = \int_{t_1}^{t_2} F dt \quad \text{impulse.}$$

From Newton's third law, we have, $F_{ab} = -F_{ba}$

$$\Rightarrow P_a(t_2) - P_a(t_1) = \int_{t_1}^{t_2} F_{ab} dt = - \int_{t_1}^{t_2} F_{ba} dt = -(P_b(t_2) - P_b(t_1))$$

$$\Rightarrow P_a(t_2) + P_b(t_2) = P_a(t_1) + P_b(t_1)$$

Problem 2) B is fastest because snow is thrown in the opposite direction.

Problem 3) a) Rocket loses mass at a rate of $\frac{dm}{dt}$. The total momentum of the system must remain constant.

$$(m+dm)v = m(v+dv) + dm(v-u)$$

$$mv + vdm = mv + m dv + vdm - u dm \Rightarrow \frac{dv}{u} = \frac{-1}{m} dm$$

$$\Rightarrow v \Big|_0^{v(t)} = -u \log m \Big|_M^{m(t)} \Rightarrow v(t) = u \log \left(\frac{M}{m(t)} \right)$$

b) put rockets on top of each other and ditch each rocket without fuel.

Problem 4) $\vec{u} = \frac{\vec{P}}{M}$, where P is the total momentum of the system as seen from the lab frame. $\Rightarrow \vec{u} = \frac{\sum m_i \vec{v}_i}{\sum m_i}$

$$\text{It's easy to see that } \vec{P}_{cm} = \sum m_i \vec{v}_i = \sum m_i (\vec{v}_i - \frac{\vec{P}}{M}) = \vec{P} - \vec{P} = 0 \checkmark$$

Similarly, the CM origin would have to be,

$$\vec{R}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$$

$$\text{It follows that, } M \vec{a}_{cm} = M \ddot{\vec{R}}_{cm} = \sum m_i \ddot{\vec{r}}_i = \sum \vec{F}_i = \vec{F}_{total}$$

Problem 5) $\vec{P}_{lab} = m\vec{v} \Rightarrow \vec{u} = \frac{\vec{P}_{lab}}{M} = \frac{m\vec{v}}{m+M}$

$$\Rightarrow v'_m = v_m - u = \frac{Mv}{m+M}; v'_M = v_M - u = 0 - u = -\frac{mv}{m+M}$$

From the CM frame definition we have: $\vec{v}_i = \vec{v}'_i + \vec{u}$

Because $P' = 0$, the conservation of momentum and energy require that the speeds in CM frame must simply reverse, otherwise any increase or decrease in speed magnitudes would violate the conservation of energy.

$$\Rightarrow v'_m = -\frac{Mv}{m+M}; v'_M = \frac{mv}{m+M} \quad \text{after collision.}$$

$$v_i = v'_i + u \Rightarrow v_m = \frac{(m-M)v}{m+M}; v_M = \frac{2mv}{m+M} \quad \text{after collision. Do the limit cases make sense?}$$

Problem 6) $T_{cm} = \frac{1}{2} \sum m_i |\vec{v}'_i|^2$; $\vec{v}_i = \vec{v}'_i + \vec{u}$

$$T = \frac{1}{2} \sum m_i |\vec{v}_i|^2$$

$$\Rightarrow T = \frac{1}{2} \sum m_i (|\vec{v}'_i|^2 + 2 \vec{v}'_i \cdot \vec{u} + |\vec{u}|^2)$$

$$= \frac{1}{2} \sum m_i |\vec{v}'_i|^2 + u \cdot \left(\sum m_i \vec{v}'_i \right) + \frac{1}{2} |\vec{u}|^2 \sum m_i$$

$$= T_{cm} + 0 + \frac{1}{2} M u^2$$

Thus, for any non-CM frame $T > T_{cm}$

Problem 7) The momentum and energy conservation laws yield:

$$\begin{cases} mv + 0 = mv_f + MV_f \\ \frac{1}{2}mv^2 + 0 = \frac{1}{2}mv_f^2 + \frac{1}{2}MV_f^2 \end{cases} \Rightarrow mv^2 = mv_f^2 + M \frac{m^2(v-v_f)^2}{m^2}$$

$$\Rightarrow 0 = (m+M)v_f^2 - 2mvv_f + (m-M)v^2 \Rightarrow ((m+M)v_f - (m-M)v)(v_f - v)$$

$$\Rightarrow v_f = \frac{(m-M)v}{m+M}; V_f = \frac{2mv}{m+M}$$

Problem 8) $\begin{cases} mv_i + MV_i = mv_f + MV_f \\ \frac{1}{2}mv_i^2 + \frac{1}{2}MV_i^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}MV_f^2 \end{cases}$

$$\Rightarrow \begin{cases} m(v_i - v_f) = M(V_f - V_i) \\ m(v_i^2 - v_f^2) = M(V_f^2 - V_i^2) \end{cases} \quad \text{divide the two eqns.}$$

$$\Rightarrow v_i + v_f = V_i + V_f \Rightarrow v_i - V_i = -(v_f - V_f)$$

Problem 9) a) From conservation laws we have:

$$\begin{cases} mv = mv_f \cos \theta + mV_f \cos \varphi \\ 0 = mv_f \sin \theta - mV_f \sin \varphi \\ \frac{1}{2}mv^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}mV_f^2 \end{cases} \quad \begin{aligned} &\text{We can eliminate } \varphi \\ &\text{to solve for } v_f \text{ \& } V_f. \\ &\text{Add the squares of the first two eqn.} \end{aligned}$$

$$v^2 - 2vv_f \cos \theta + v_f^2 = V_f^2, \text{ eliminate } V_f \text{ in (III)}$$

$$\Rightarrow v_f = v \cos \theta \Rightarrow V_f = v \sin \theta$$

$$\text{b) From (II) } \Rightarrow m(v \cos \theta) \sin \theta = m(v \sin \theta) \sin \varphi \Rightarrow \begin{cases} \cos \theta = \sin \varphi \\ \theta = 0 \text{ trivial} \end{cases}$$

$$\Rightarrow \varphi = 90 - \theta$$

Problem 10) a) $F = \frac{dP}{dt} = \frac{d(mv)}{dt} = m \frac{dv}{dt} + \frac{dm}{dt} v = 0 + \sigma v$

$$\Rightarrow F = \sigma v$$

$$\text{b) } \frac{dT}{dt} = \frac{d}{dt} \left(\frac{1}{2}mv^2 \right) = \frac{dm}{dt} \left(\frac{v^2}{2} \right) = \frac{\sigma v^2}{2}$$

$$\text{c) } \frac{dW}{dt} = \frac{F dx}{dt} = F \cdot v = \sigma v^2$$

$$\text{d) Work-energy theorem states that, heat} = W - T = \sigma v^2 - \frac{\sigma v^2}{2} = \boxed{\frac{\sigma v^2}{2}}$$

Problem 11) $\sum F = F_s - \sigma Lg = \frac{dP}{dt} = \frac{d(\sigma y \dot{y})}{dt}$

$$= \sigma y \ddot{y} + \sigma \dot{y}^2 \quad \text{I}$$

The part of chain in air is in free fall. $\Rightarrow \ddot{y} = -g \quad \text{II}$

From energy conservation we have, $y = \sqrt{2g(L-y)} \quad \text{III}$

$$\text{I II III } \Rightarrow F = \sigma Lg - \sigma y g + 2\sigma(L-y)g = 3\sigma(L-y)g$$

